AWS VPC PEERING

Private EC2 Connectivity Across Two Virtual Private Clouds

Final Technical Project Report | EWB-VPC-Peering

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AWS Region: ap-southeast-2 (Sydney)

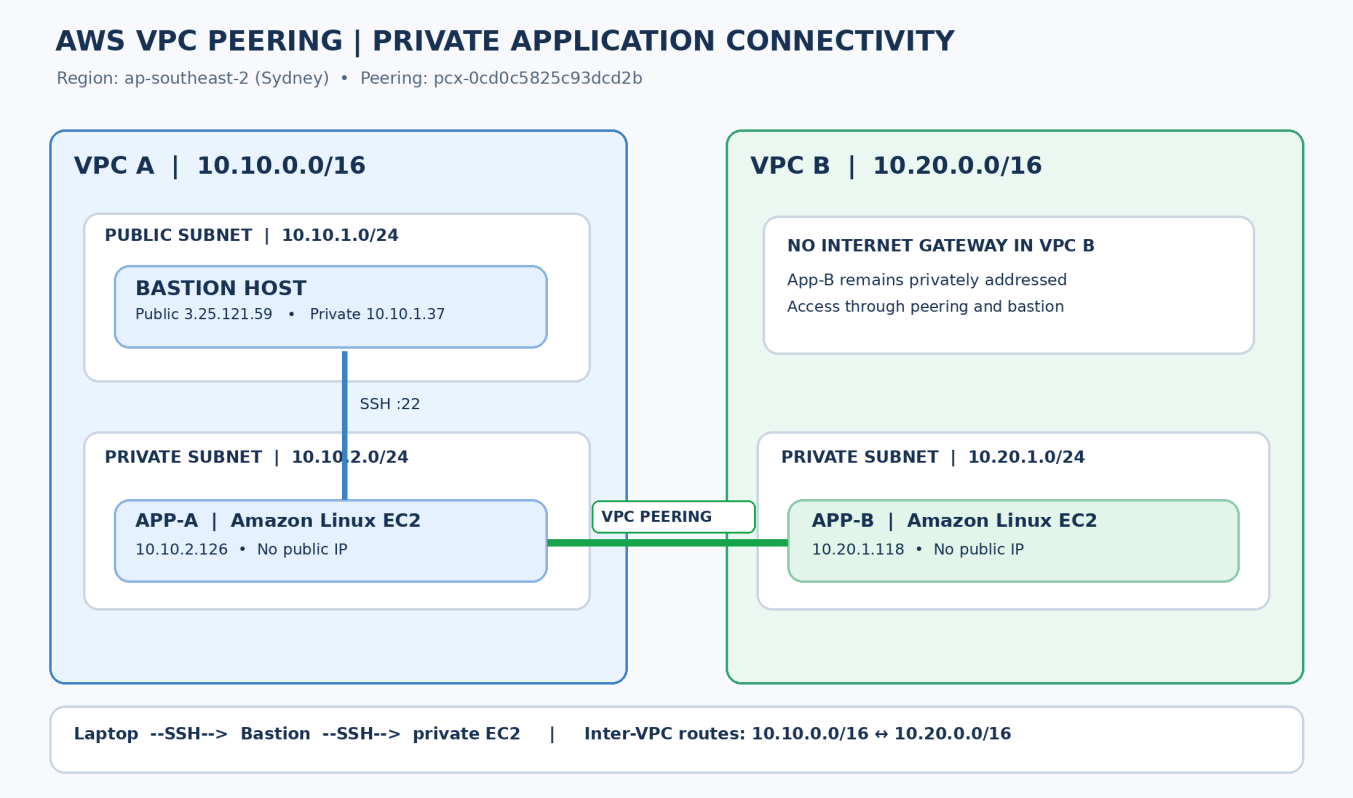


Figure 1. Designed architecture based on observed AWS resource configuration.

Executive summary. Two non-overlapping VPCs are connected through active VPC peering. A bastion host in VPC A enables administrative access to private EC2 instances. Reciprocal route-table entries and scoped security groups support private inter-VPC communication. The observed tests demonstrate successful ICMP connectivity in both directions and SSH access through the bastion.

1. Objectives and scope

Implement isolated AWS networks, establish private routing without exposing application instances to the public internet, apply security-group controls, and verify private network reachability. The screenshots were supplied as a ZIP archive and selected original captures are included in this report and repository.

2. AWS resource inventory

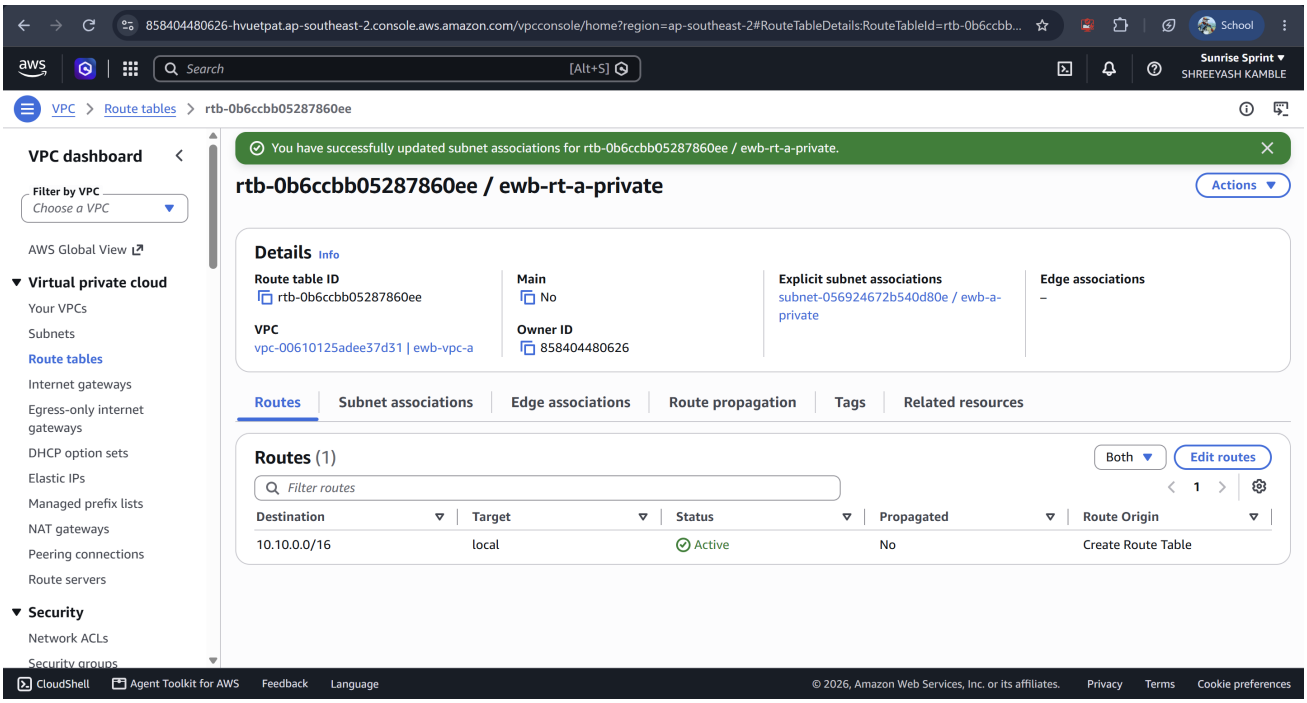
Resource	Observed configuration
VPC A	vpc-00610125adee37d31 | 10.10.0.0/16
VPC B	vpc-028e2566aa68b8e59 | 10.20.0.0/16
Peering	pcx-0cd0c5825c93dcd2b | Active
Bastion	3.25.121.59 (public); 10.10.1.37 (private)
App-A	10.10.2.126 | private EC2
App-B	10.20.1.118 | private EC2
Subnets	A public 10.10.1.0/24; A private 10.10.2.0/24; B private 10.20.1.0/24

3. Network architecture

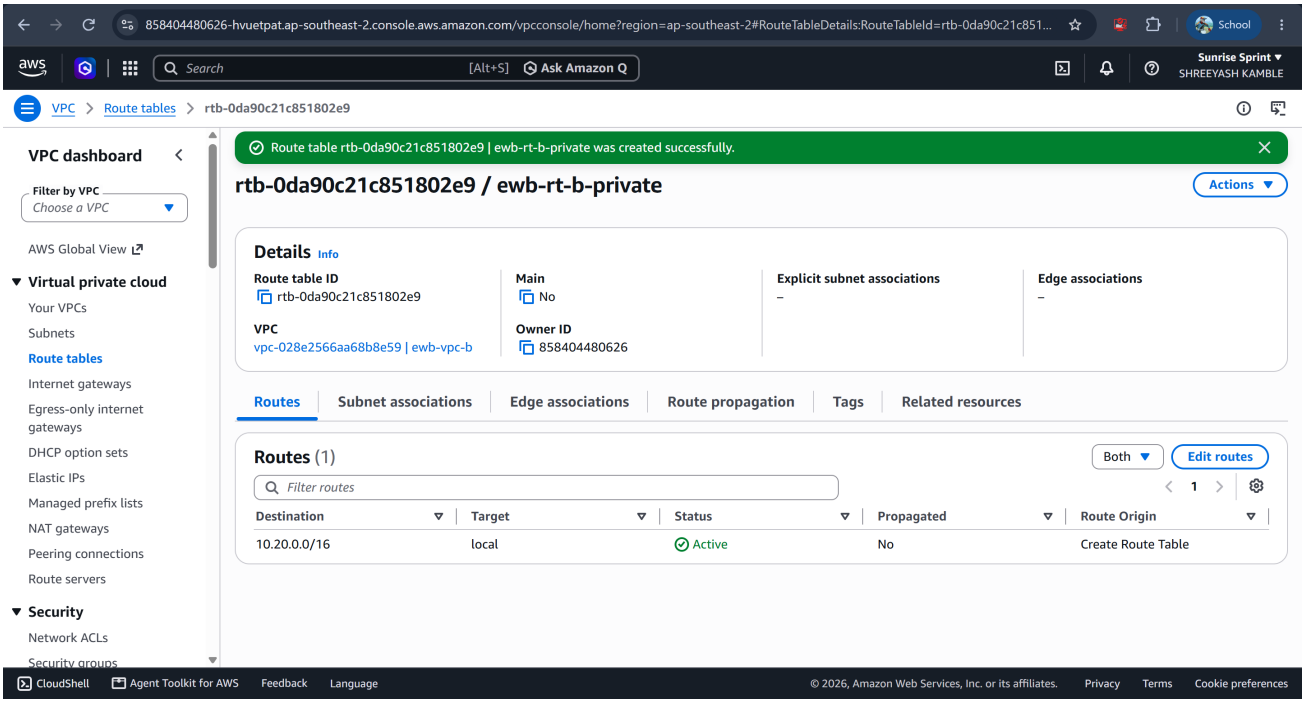
VPC A contains a public subnet hosting the bastion and a private subnet hosting App-A. VPC B contains the App-B private subnet. The peering connection is not transitive and does not automatically exchange routes; route-table entries are required on both sides.

4. Routing and subnet associations

VPC A route tables include a route for 10.20.0.0/16 targeting the VPC peering connection. VPC B private route table includes 10.10.0.0/16 targeting the same connection. The VPC A public route table also provides 0.0.0.0/0 via its internet gateway for the bastion subnet. The application subnets remain private.



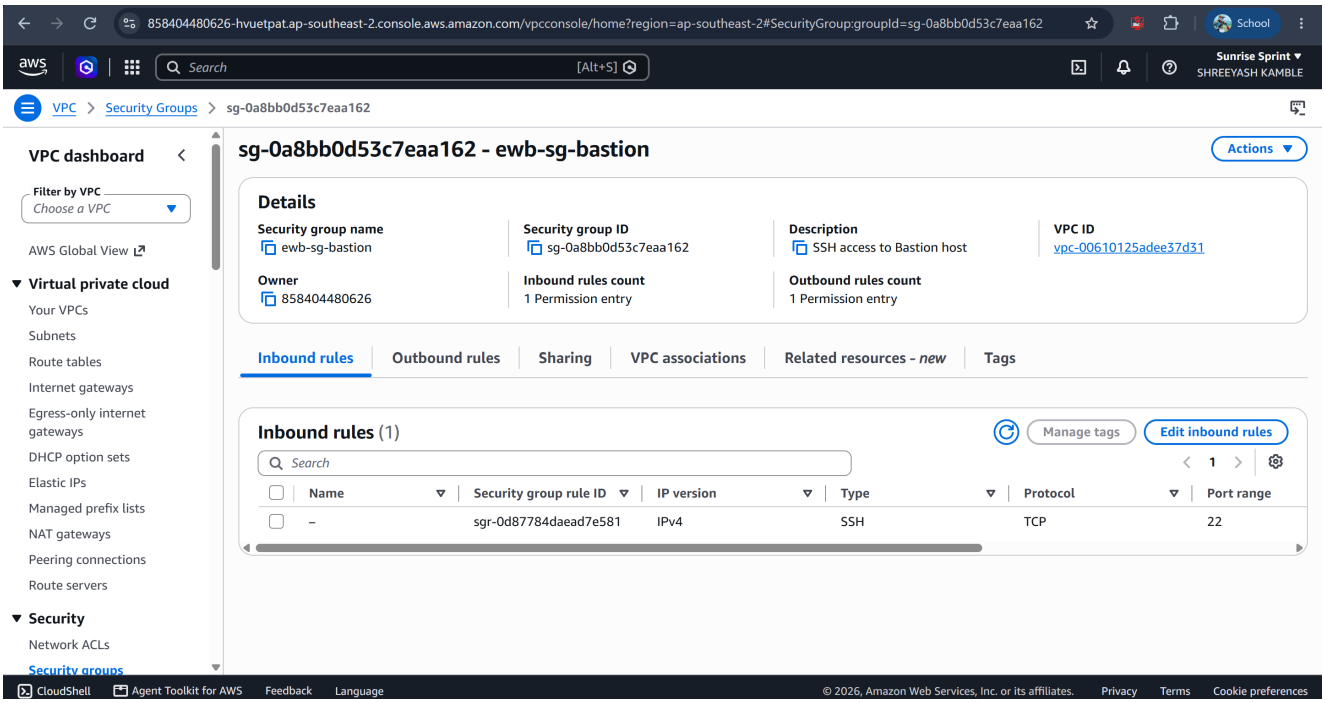
Evidence A. VPC A private route table and peering destination.



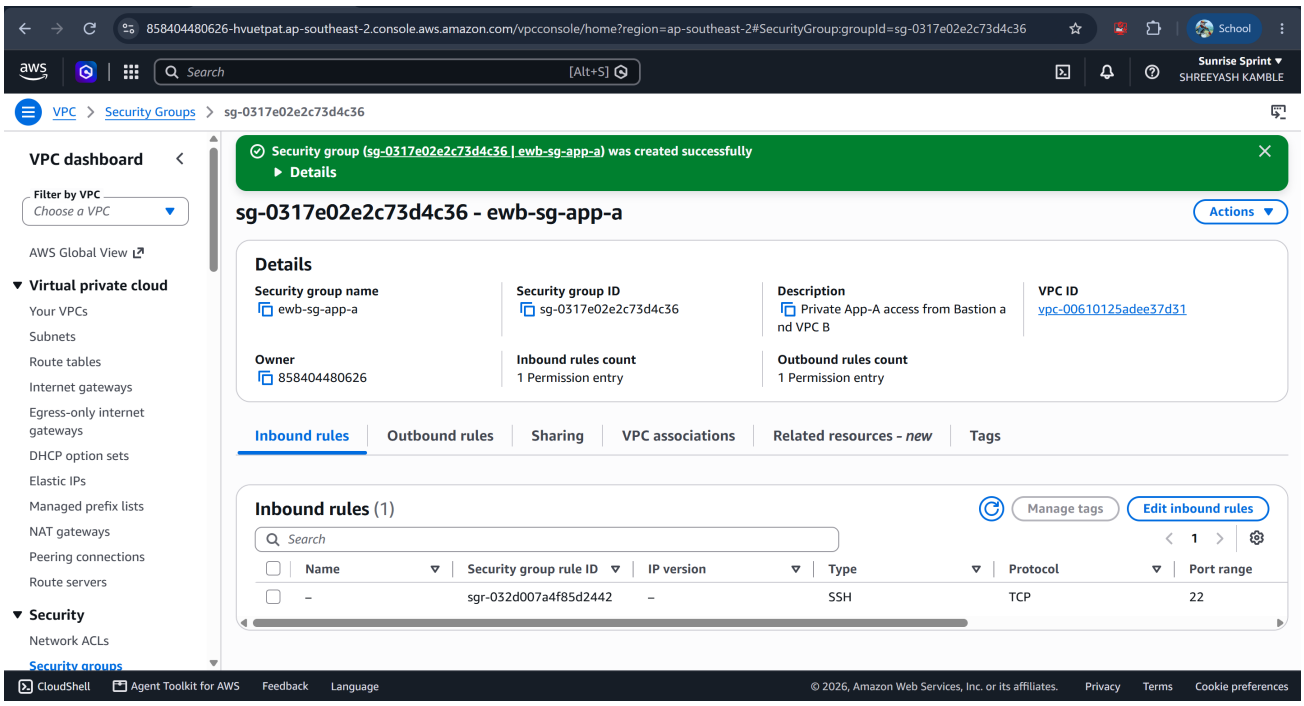
Evidence B. VPC B private route table and reciprocal route.

5. Security architecture

Security groups restrict SSH and ICMP to known private sources. App-A allows ICMP Echo Request from App-B (10.20.1.118/32). App-B allows ICMP Echo Request from App-A (10.10.2.126/32). SSH rules permit access through the bastion and scoped private hosts. The reviewed default network ACLs allow inbound and outbound traffic at rule 100; security groups provide the finer instance-level controls.



Evidence C. Bastion security-group configuration.

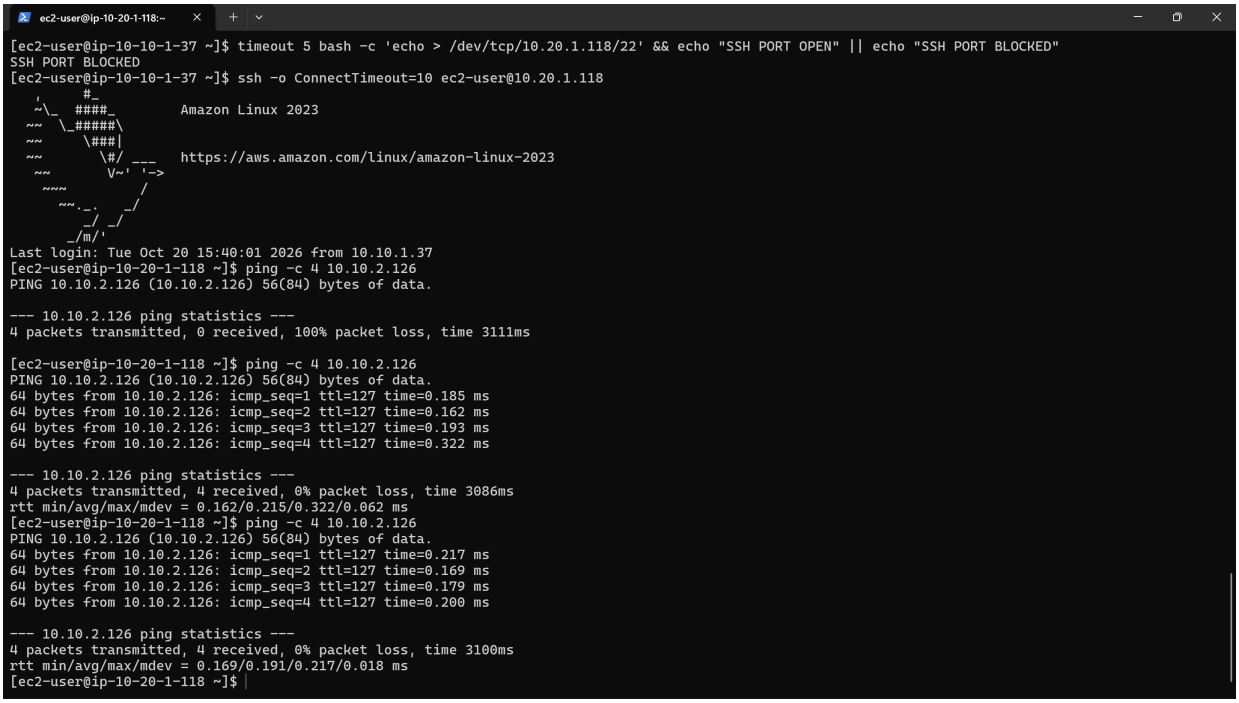


Evidence D. App-A security-group configuration.

6. Connectivity verification

Test	Observed result
Local workstation to Bastion	SSH login succeeded
Bastion to App-B	SSH login succeeded
App-A to App-B ICMP	4 transmitted, 4 received; 0% loss
App-B to App-A ICMP	4 transmitted, 4 received; 0% loss

The latest supplied App-B terminal screenshot confirms a successful reverse ping to 10.10.2.126 with an average round-trip time of approximately 0.193 ms. Successful bidirectional ICMP is consistent with correct peering routes and inbound security rules.



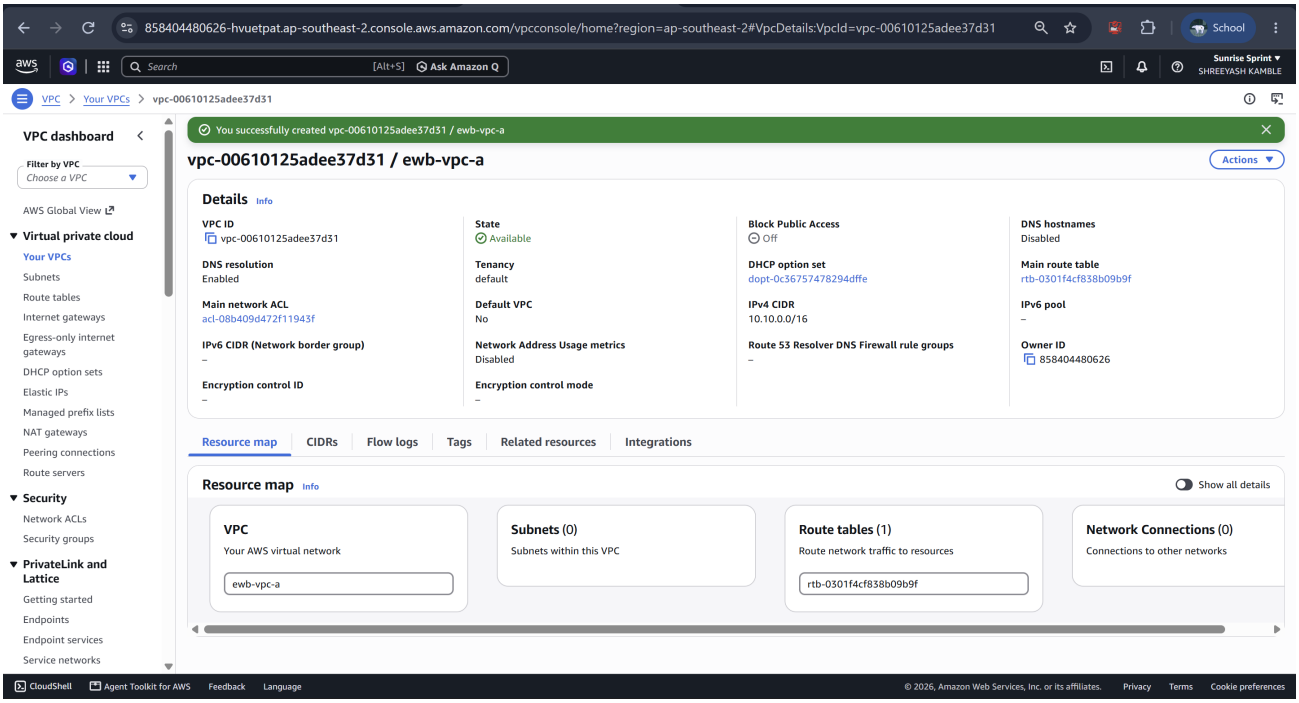
Evidence E. Original terminal capture from the supplied archive.

7. Controlled failure and recovery

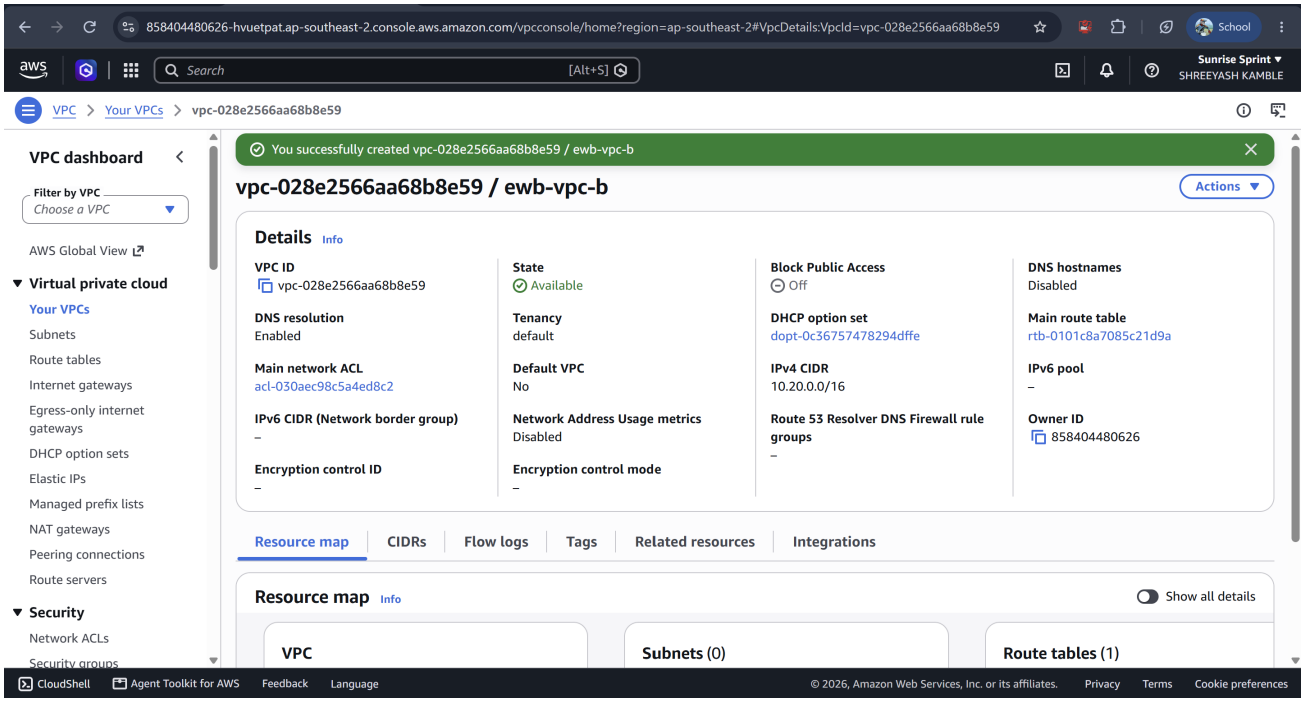
Test procedure: remove only the App-B inbound ICMP Echo Request rule permitting source 10.10.2.126/32; run ping from App-A to 10.20.1.118 and record failure; restore the same ICMP rule and rerun the test. This demonstrates the effect of instance-level security policy without changing VPC peering or routing.

Evidence note: The controlled-failure and recovery sequence was reported as completed in the preceding project work. The present curated screenshots are not labeled as a definitive before/blocked/restored three-image sequence, so this report does not claim that each phase has been independently re-verified from the archive. Add clearly labeled captures before academic submission if your evaluator requires that sequence.

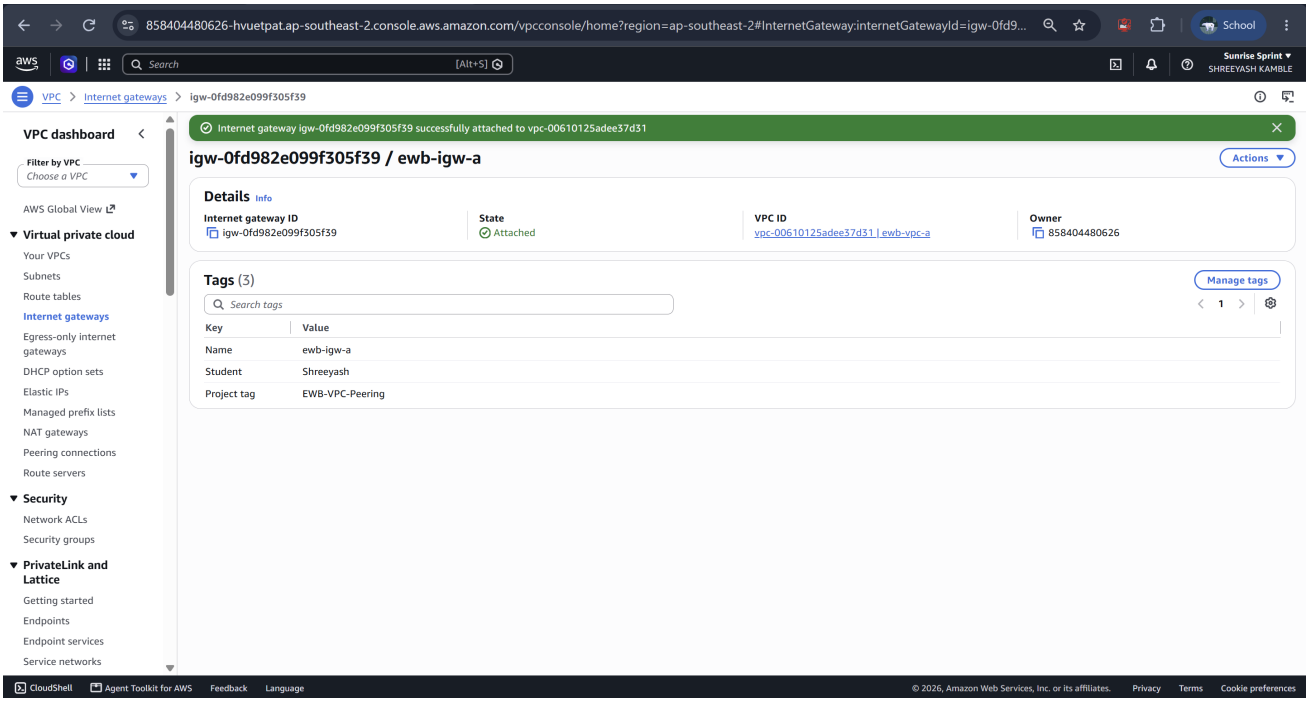
8. Implementation evidence



Evidence F. VPC A creation.



Evidence G. VPC B creation.



Evidence H. Internet gateway attached to VPC A.

9. Findings and conclusions

The infrastructure demonstrates private, bidirectional communication across two separate VPC CIDR ranges using VPC peering. Private EC2 instances have no public IPv4 addresses, while the bastion serves as a controlled administrative entry point. Successful SSH and ping tests support the functional result. Security-group changes can selectively disrupt ICMP while leaving the peering relationship intact.

10. Submission and operational notes

This report, the high-resolution architecture diagram, and selected original screenshots are included in the accompanying GitHub-ready ZIP. Do not publish SSH private keys, credentials, account passwords, or secrets. Before terminating resources, retain the final test screenshots and verify whether the assignment expects cleanup evidence. Once grading no longer requires the environment, terminate chargeable resources and remove unused networking resources to avoid ongoing charges.

Evidence limitation: Screenshots capture AWS configuration at a point in time. They do not prove the current state of the AWS account after the capture date. No live AWS account access was used to prepare this report.

Appendix: Key resource IDs

Peering: pcx-0cd0c5825c93dcd2b
App-A security group: sg-0317e02e2c73d4c36
App-B security group: sg-0b7da40e7bfb6594f
Bastion security group: sg-0a8bb0d53c7eaa162
Internet gateway: igw-0fd982e099f305f39